
The Rust programming language



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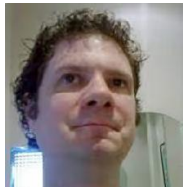
What is Rust?

- ▶ **Systems programming language** focused on three goals at once:
 - + **safety** : no null-pointer dereferences, no data races, no use-after-free,
 - + **speed** : performance on par with C and C++,
 - + **concurrency** : “fearless” parallel programming.
- ▶ Compiled and statically typed, with **no garbage collector**.
- ▶ Memory safety is guaranteed at compile time by the **ownership** and **borrow-checking** model.

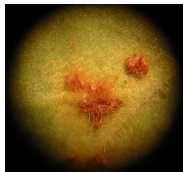
Slogan: *“A language empowering everyone to build reliable and efficient software.”*

Where does the language come from?

- ▶ Started in 2006 as a personal side project of **Graydon Hoare**, an engineer at Mozilla.
- ▶ The name suggests robustness, evoking the *rust fungi* that are known for being extremely resilient.
- ▶ Grew through a large open-source community; today it is stewarded by the independent, non-profit **Rust Foundation** (founded 2021).
- ▶ For many years running, the *most admired* language in the Stack Overflow Developer Survey.*



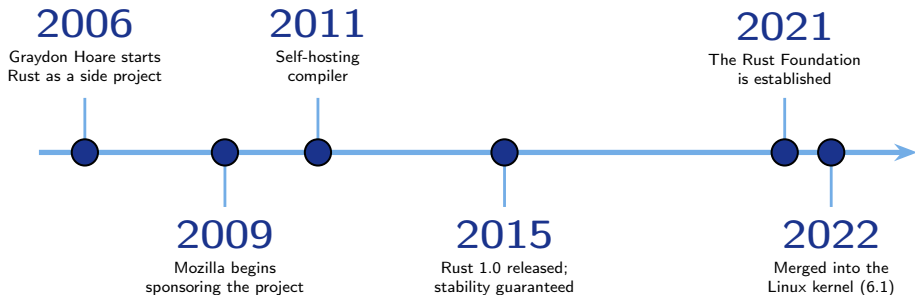
Graydon Hoare,
original author of Rust



Rust fungus

*<https://survey.stackoverflow.co/2024/technology>

History



Characteristics of Rust

- 😊 **memory safety** without a garbage collector
- 😊 strong **compile-time checks** : “if it compiles, it usually works”
- 😊 **performance** comparable to C / C++
- 😊 fearless **concurrency** : data races are caught at compile time
- 😊 **modern tooling** : [cargo](#), [crates.io](#)
- 😊 **expressive** language elements, leading to concise code
- 😊 **free** , open source, and platform independent
- 😞 steeper learning curve (especially the borrow checker)
- 😞 longer compile time

AI and Rust: A Powerful Duo



A first taste

```
fn main() {  
    let name = "Rustacean".to_string();  
    let who = name; // ownership moves to `who`  
    println!("Hello, {who}!");  
}
```

Rustacean [ruh-STAY-shun] is a member of the Rust community. The word originates from the blend of *Rust* and *crustacean* (the taxonomic group of Ferris 🦀 the crab).

Development environment

- ▶ In this course, we will use **VSCodium** (<https://vscodium.com/>).
- ▶ It is a **free / libre, open-source build of Visual Studio Code**, i.e. the same editor, but:
 - built **without Microsoft's telemetry and tracking**,
 - released under the MIT license, community-driven binaries.
- ▶ We will also install the **rust-analyzer** extension:
 - real-time type information and inline error highlighting,
 - autocompletion, go-to-definition, refactoring,
 - surfaces the compiler's checks **as you type**, exactly the feedback loop that makes Rust so reliable.

